Started on	Wednesday, 30 September 2026, 9:56 PM
State	Finished
Completed on	Wednesday, 30 September 2026, 10:05 PM
Time taken	9 mins 10 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
#include<stdlib.h>
int findFloor(int arr[], int low, int high, int x){
    int floor_val = -1;
    while(low<=high){
        int mid = low + (high - low)/2;
        if(arr[mid] == x){
            return arr[mid];
        }
        else if(arr[mid]<x){
            floor_val = arr[mid];
            low = mid + 1;
        }
        else{
            high = mid - 1;
        }
    }
    return floor_val;
}
int main(){
    int n,x;
    if(scanf("%d", &n) != 1) return 0;
    int *arr = (int *)malloc(n*sizeof(int));
    for(int i = 0;i<n;i++){
        if(scanf("%d",&arr[i])!=1) return 0;
    }
    if(scanf("%d",&x)!=1) return 0;
```

```
28      int result = findFloor(arr, 0, n-1, x);
29      printf("%d\n", result);
30
31      free(arr);
32      return 0;
33  }
```

	Input	Expected	Got	
✔	6 1 2 8 10 12 19 5	2	2	✔
✔	5 10 22 85 108 129 100	85	85	✔
✔	7 3 5 7 9 11 13 15 10	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.